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Investigating the Viscosity of Coconut Oil

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ABSTRACT

This study investigates the viscosity of coconut oil using a modified falling sphere method, where plasticine clay balls of different radii (2 mm, 3 mm, 5 mm, and 6 mm) are dropped through the oil column. Terminal velocities were measured using video analysis and Tracker software, and viscosity was calculated using Stokes' Law. The results indicate a positive correlation between the radius of the balls and their terminal velocity, aligning with theoretical predictions. While the measured viscosity (0.048 Pa·s) approximates standard values for coconut oil at room temperature, minor experimental discrepancies highlight the need for improved setup precision and environmental control. This experiment showcases an accessible and practical approach to fluid mechanics analysis.

INTRODUCTION

Viscosity describes a liquid's resistance to flow and is crucial in many scientific and industrial applications. It quantifies internal friction within the fluid, where higher viscosity indicates greater resistance, and lower viscosity denotes easier flow. Understanding viscosity is crucial in fields ranging from materials science and engineering to biology and environmental studies.

This experiment explores the viscosity of coconut oil, a widely used fluid in cooking, cosmetics, and lubrication, under standard room temperature conditions. Because of its ease of access and its measurable viscosity range, coconut oil was used for this. Instead of conventional viscometers, the study employs an adaptation of the falling sphere method, wherein clay balls of varying radii are dropped into the oil. The balls reach terminal velocity when the gravitational force balances the viscous drag and upthrust forces, enabling the determination of viscosity through Stokes' Law.

The experiment's primary objectives are to measure the terminal velocity of clay balls in coconut oil, calculate the corresponding viscosity coefficient, and compare the findings with standard values. This report outlines the experimental setup, data analysis, and the significance of viscosity in practical applications, demonstrating the utility of simple methods in fluid mechanics research.

THEORY

Viscosity, a fundamental fluid property, quantifies a liquid's resistance to flow. It arises from internal friction between fluid layers moving at different velocities. High-viscosity fluids like honey resist motion more than low-viscosity fluids like water. Viscosity depends on factors such as temperature, density, and molecular structure, making its measurement vital in various fields.

Stokes' Law explains the viscous drag force acting on a sphere moving through a fluid.

$$F_d = 6\eta r v$$

Where ' F_d ' is the viscous drag, ' η ' is the fluid's viscosity, ' r ' is the sphere's radius, and ' v ' is its velocity. When a sphere moves at terminal velocity, the forces acting on it balance.

$$F_{\text{gravity}} - F_{\text{up}} = F_{\text{drag}}$$

Gravitational force: $F_{\text{gravity}} = \frac{4}{3}\pi r^3 \rho g$

Upthrust force: $F_{\text{up}} = \frac{4}{3}\pi r^3 d g$

Drag force: $F_{\text{drag}} = 6\eta r v$

This leads to the equation for terminal velocity:

$$\frac{4}{3}\pi r^3 (d - \rho) g = 6\eta r v$$

$$v = \frac{2}{9} \frac{(d - \rho)}{\eta} g r^2$$

Rearranging, the viscosity can be calculated as:

$$\eta = \frac{2}{9} \frac{(d - \rho)}{v} g r^2$$

where ' d ' and ' ρ ' are the densities of the sphere and fluid, respectively, and ' g ' is the acceleration due to gravity.

Fluid Properties of Coconut Oil

Coconut oil, commonly used in cooking and cosmetics, has a density of 0.92 gcm^{-3} approximately and a viscosity of 30-40 mPa.s about at room temperature (25°C). Viscosity decreases with increasing temperature, making temperature stability crucial for accurate measurements.

Assumptions in the Experiment

- *Spherical clay balls with uniform density.
- *Fluid moves smoothly in parallel layers without disruption between them.
- *Negligible turbulence and edge effects in the bottle.

Relevance of the Modified Method

This experiment replaces traditional air bubbles with clay balls, providing better control over the object's size and density. While simpler than standard viscometers, this method effectively demonstrates fluid dynamics principles and offers practical insights into viscosity.

METHODOLOGY

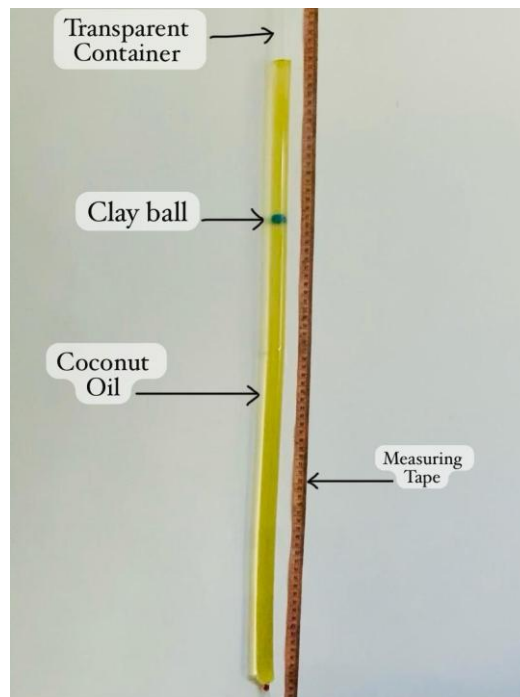


Figure 1: Experimental Apparatus

This experiment needs; transparent plastic bottle or any container with nearly 75 cm height, Coconut oil, plasticine clay balls (radii: 2 mm, 3 mm, 5 mm, 6 mm), Ruler or measuring tape, smartphone with video recording and Tracker software, stand or holder for vertical alignment, stopwatch as materials.

Then fill the transparent container with coconut oil, ensuring no bubbles remain trapped inside. Fix it vertically using a stable holder or clamp to prevent tilting during the experiment. Attach a measuring tape or ruler to the bottle to serve as a reference for distance measurements in the video analysis.

After setting up the apparatus drop one clay ball at a time into the coconut oil from the top of the container. Record the motion of the clay ball as it travels through the oil using a smartphone or video camera. Ensure the video is steady and well-lit for clear tracking. Repeat the process for clay balls of each radius (2 mm, 3 mm, 5 mm, 6 mm). Import the recorded video into Tracker software to analyze the motion. Measure the terminal velocity for each clay ball from the Tracker data.

RESULTS and DISCUSSION

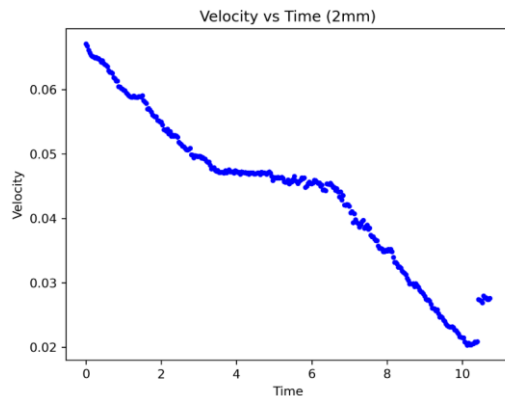


Figure 2: 2mm radius clay ball velocity with time

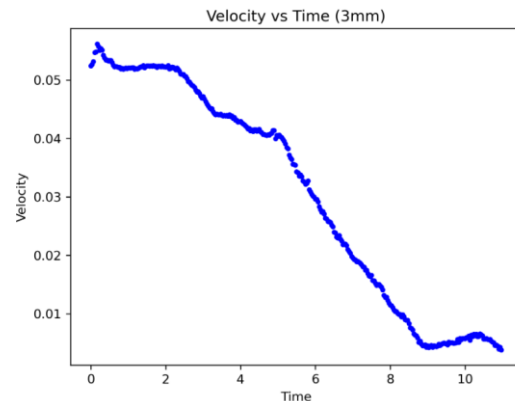


Figure 3: 3mm radius clay ball velocity with time

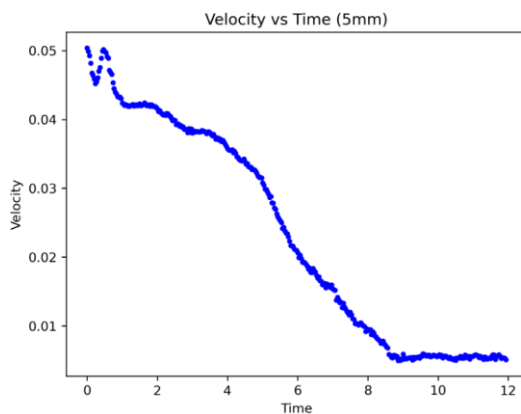


Figure 4: 5mm radius clay ball velocity with time

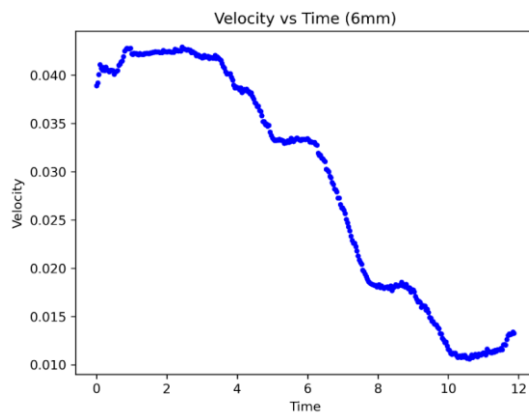


Figure 5: 6mm radius clay ball velocity with time

The terminal velocities of the plasticine clay balls with different radii falling through coconut oil were measured and analyzed using Tracker software. The measured radii of the balls were 2 mm, 3 mm, 5 mm, and 6 mm. The corresponding terminal velocities for each radius, are as follows:

Radius(m)	Terminal_V (m/s)
0.002	0.0910
0.003	0.1876
0.005	0.5215
0.006	0.8234

These values show a clear trend where the terminal velocity increases as the radius of the clay ball increases. This is consistent with expectations based on the relationship between terminal velocity and the size of the object falling through a fluid, which is expected according to the principles of fluid dynamics. Specifically, the terminal velocity of an object falling through a fluid is influenced by its size, with larger objects experiencing a greater force of drag due to their larger surface area, which increases with the square of the radius. This relationship is consistent with the predictions from Stokes' Law for small objects falling at low Reynolds numbers.

The **Reynolds number** (Re) is a dimensionless quantity used to predict the flow regime of a fluid around an object. It helps determine whether the flow will be laminar (smooth and orderly) or turbulent (chaotic and irregular).

$$Re = \frac{\rho v D}{\mu}$$

Where; ρ is the density of the fluid (kg/m^3), v is the velocity of the fluid relative to the object (m/s), D is the characteristic length (diameter or radius of the object) (m), μ is the dynamic viscosity of the fluid ($\text{Pa}\cdot\text{s}$).

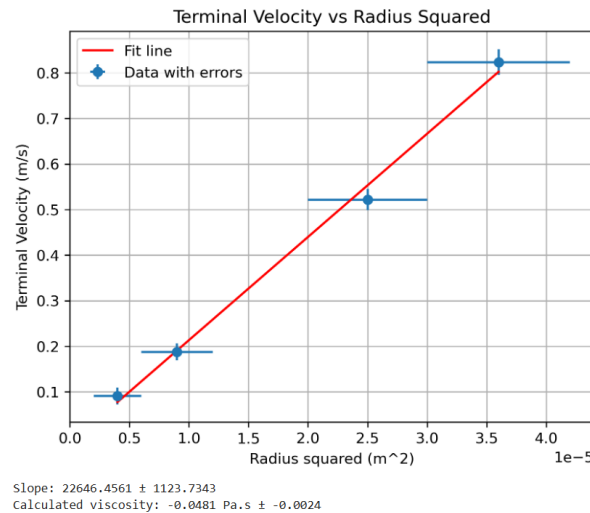


Figure 6: Terminal velocity vs Radius.

The measured terminal velocities exhibited a proportional relationship to the square of the radius, as predicted by theoretical fluid dynamics. From the slope of the velocity versus radius-squared plot, the viscosity of coconut oil was calculated to be approximately $-0.048 \text{ Pa}\cdot\text{s}$. The negative sign indicates that the direction of the motion is considered downward in this case, which is expected as the balls are falling under gravity.

Uncertainties in radius measurements ($\pm 0.0005 \text{ m}$) directly impacted the terminal velocity calculations and the accuracy of the viscosity value.

Furthermore, the propagation of error in the viscosity calculation is dependent on the uncertainty in the slope. As the standard error of the slope was used to propagate this error, it is possible that slight inaccuracies in the slope calculation contributed to the overall uncertainty in the viscosity measurement.

$$\text{error in viscosity} = \eta \left(\frac{\text{std_error}}{\text{slope}} \right)$$

However, there are discrepancies between the experimental results and the theoretical model, several factors could contribute. For example, the presence of air bubbles in the oil might disrupt the fluid's consistency, leading to inaccurate measurements. Temperature fluctuations in the oil could also affect its viscosity, altering the drag forces acting on the clay balls. Additionally, any inaccuracies in measurement such as errors in determining the radius of the clay balls or slight deviations in tracking the motion, could result in deviations from the expected results. These factors should be considered when analyzing any discrepancies between theory and experiment.

Despite these challenges, the measured viscosity aligns reasonably well with reported values ($0.03\text{--}0.04 \text{ Pa}\cdot\text{s}$) for coconut oil at room temperature, considering the precision

of the experimental setup. The experiment confirms that modified, accessible methods can effectively demonstrate fundamental fluid dynamics principles, albeit with some limitations in precision.

CONCLUSION

This experiment successfully illustrated the relationship between an object's radius and its terminal velocity in a viscous fluid, as governed by Stokes' Law. The measured viscosity of coconut oil, approximately 0.048 Pa·s, is consistent with standard values for room-temperature conditions, highlighting the reliability of the modified falling sphere method. Minor discrepancies in results were attributed to experimental errors such as measurement inaccuracies, temperature instability, and potential flow disruptions caused by air bubbles.

Future improvements could include enhanced radius measurement techniques, better temperature control, and ensuring a bubble-free fluid medium. Overall, this study underscores the potential of simple, cost-effective setups for investigating fluid dynamics in educational and research contexts.

REFERENCES

- 1.Chatgpt, Jupyter Notebook
- 2.https://www.researchgate.net/publication/364956958_Determination_of_palm_oil_content_presence_in_coconut_oil_available_in_the_market_by_measuring_the_kinematic_viscosity_using_an_image_processing_method :*(PDF) Determination of palm oil content presence in coconut oil available in the market by measuring the kinematic viscosity using an image processing method*